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A-A-59407
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COMMERCIAL ITEM DESCRIPTION

TEST SET GROUP, RADIO

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

This Commercial Item Description (CID) outlines the minimum technical and performance requirements for a Radio Test Set Group used to verify, test, and repair Life Support survival radios such as the AN/PRC-112, AN/PRC-90-2, AN/URT-33, and related communications and electronic equipment. The test set shall be an integrated, self-contained, portable unit capable of functioning as a spectrum analyzer, receiver, transmitter, oscilloscope, digital multimeter, and signal generator to allow trouble shooting and operation verification of communications and electronic equipment. The functional testing of radios and beacon shall be automated to include full functional testing and user selectable single function tests for each radio or beacon to verify operation and performance within specified limits.

2. SALIENT CHARACTERISTICS.

2.1 Class 3. This test set is categorized as Class 3 test equipment in the latest version of MIL-PRF-28800 “Performance Specification, Test Equipment for Use with Electrical and Electronic Equipment”.

2.2 Safety. This test set shall meet all safety requirements as specified for Class 3 test equipment in MIL-PRF-28800. For safety ground, all accessible surfaces of the equipment shall be at ground potential. The power cable shall include a safety ground conductor. Disconnecting of safety ground shall also disconnect the power conductors.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document may be sent to: 311 HSW/YACL, 8107 13th Street, Brooks AFB, TX 78235 on the enclosed DD Form 1426, Standardization Document Improvement Proposal, appearing at the end of this document or by letter.

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2.3 Equipment components. Parts, materials, and processes concerning electronic components shall be as specified in paragraphs 3.2 through 3.2.3.3 of MIL-PRF-28800.

2.4 Functional requirements. The test set shall comply with all applicable requirements of Class 3 test equipment in accordance with section 3.3 of MIL-PRF-28800 excluding requirements for internal cooling, front panel display indicators, and digital interface which are specified below:

2.4.1 Internal cooling. The test set shall incorporate internal cooling as required to comply with the high temperature operation requirement within this specification.

2.4.2 Front panel display indicators. The test set shall be configured with a color CRT to provide information to the user when operating the test set. This includes all modes of operation: maintenance, calibration, self-test, debug, test limit and condition setting, etc. Minimum display dimensions are 5.2 inches wide x 5.6 inches high with a resolution of 640 pixels x 350 lines or greater. The test set display shall be positioned to provide easy operator viewing of information during test set operation.

2.4.3 Digital interface. The test set shall have the capability for remote operation using external controllers and video monitors. This capability shall include the capability for data/information transmission to remote locations over the local area network. The interface for remote operation shall comply with ANSI/IEEE-STD-488.2. The test set shall be configured with a 9-pin, sub-miniature D, male RS-232 connector to allow use of other serial devices. The test set shall be configured with a 9-pin, subminiature D, female EGA video connector to allow use of an external monitor. The test set shall also be configured with an IEEE-488, 24-pin connector to allow use of a printer.

2.5 Electrical power sources and connections. The test set shall operate from nominal commercial, military, and shipboard power sources operating at AC voltages of 95 to 130 VRMS (50 to 400 Hertz) and 190 to 260 VRMS (50 to 400 Hertz). The test set shall also be capable of operating with an external DC source of 22 to 30 VDC.

2.5.1 Power switch and indicator. The test set shall be configured with an on/off power switch. The switch shall be located on the exterior of the test set to provide operator access. A visual indicator shall be provided at the front of the test set to indicate when the test set is energized.

2.5.2 Power selection switch. When the configuration of the test set requires an AC power selection switch, a safety mechanism shall be incorporated to prevent accidental switching or connection of mismatched AC voltages.

2.5.3 Interruption of power source. Voltage or frequency interruptions shall not cause damage to the test set. A power source interruption is when the voltage falls in the range of 0 to 70 percent of nominal or frequency falls in the range of 0 to 90 percent of nominal. Test set performance is not required during interruption of the power source. After power interruptions of any duration, the test

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set shall automatically resume operation at either the last operating condition before interruption, or a default power up condition.

2.6 Physical Characteristics.

2.6.1 Mechanical stability. The test set shall be mechanically stable when positioned in its normal operational configuration on a bench to preclude tipping during normal handling and operation.

2.6.2 Dimensions. The test set shall fit within a 13 inches high x 21 inches wide x 27 inches deep volume.

2.6.3 Weight. The weight test set shall not exceed 85 pounds.

2.7 Storage and transit case. A durable storage and transit case shall be provided. This case shall serve as an enclosure for the test set, its accessories, and operating instructions. Stowage shall not adversely affect the carrying stability of the test set and shall be contained in a manner that prevents damage to the accessories or test set during transit. Handles shall be provided and positioned to ensure carrying stability. All hard-edged corners shall be designed to preclude injury to personnel. Airtight transit cases shall have a mechanism for equalizing air pressure within the enclosure when the enclosure is sealed.

2.8 Environmental requirements. In accordance with MIL-PRF-28800, the test set shall withstand and operate under the following environmental conditions for Class 3 test equipment:

2.8.1 Temperature, non-operating. The test set shall conform to the specified performance and accuracy after being stored in the temperature range of minus 40 degrees to plus 71 degrees Celsius (C.). Testing shall be in accordance with 3.8.2.1.2 and 4.5.5.1 of MIL-PRF-28800.

2.8.2 Temperature operating. The test set shall conform to the specified performance and accuracy while being operated in a temperature range of 0 to plus 50 degrees Celsius (C). Testing shall be in accordance with 3.8.2.2 and 4.5.5.1 of MIL-PRF-28800.

2.8.3 Relative humidity. The test set shall conform to the specified performance and accuracy while exposed to environments with of 5 to 95 percent relative humidity. Testing shall be in accordance with paragraphs 3.8.2.3 and 4.5.5.1 of MIL-PRF-28800.

2.8.4 Vibration. The test set shall conform to the specified performance and accuracy random vibration levels of 5 to 500 Hz. Testing shall be in accordance with paragraphs 3.8.4.1 and 4.5.5.3.1 of MIL-PRF-28800.

2.8.5 Bench handling. This equipment shall be able to withstand shipping and handling occurring in the work area. Testing shall be in accordance with paragraphs 3.8.5.3 and 4.5.5.4.3 of MIL-PRF-28800.

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2.8.6 Shock. The test set shall meet its performance and accuracy requirements in an operating environment after being subjected to a 30g half-sine shock. Testing shall be in accordance with paragraphs 3.8.5.1 and 4.5.5.4.1 of MIL-PRF-28800.

2.9 Calibration. Calibration adjustments shall be accomplished by variable value components which are adjustable by the use of simple means. Any calibration by substitution of selected components or parts is unacceptable unless specifically approved in advance. The calibration adjustments whenever possible shall be accessible without removal of the instrument case or modules. The manufacturer shall provide all procedures, special tools, and instructions. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours. Changes in environmental conditions within the operating range of the test set will not require additional calibration except for internal calibration procedures and signal path compensation to meet performance requirements.

2.10 Electromagnetic Interference (EMI) and Electrostatic Discharge (ESD) control. The test set shall operate in the intended environment unaffected by Electromagnetic Interference (EMI) as stated in MIL-PRF-28800, paragraphs 3.9, 4.5.6.5 and Table 6, and comply with performance control requirements concerning Electrostatic Discharge sensitivity (ESD) as outlined in MIL-STD-1686C.

2.11 Panel markings, warning label, and product identification.

2.11.1 Panel markings. Each test set shall be provided with panel markings which enable the operator to identify the functions and use of all items requiring operator intervention.

2.11.2 Warning label. A warning label shall provide information about the possible location, nature and the extent of a hazard. Warning markings shall have a high contrast between the letter and background colors. The markings shall be as permanent as the normal life expectancy of the test equipment and located as close as possible to the point of danger. Warning markings shall be in accordance with the requirements outlined in ANSI/ISA-S82.01-1994 or ANSI/ISA-S82.02-1994, or both as applicable.

2.11.3 Identification plates. Two identification plates shall be affixed to the test set; one shall be affixed to the tester and the other shall be mounted outside the transit case in a readable position. Each shall contain the following information in legible and permanent print:

a. Equipment name;
b. Manufacturer’s model and/or serial number;
c. National Stock Number (NSN) (available after stocklist) and
d. Contract Number

2.12 Reliability tests. Reliability verification shall conform with paragraphs 3.13 and 4.5.8.1 of MIL-PRF-28800. The equipment shall operate for 2,000 hours without failure with a statistical certainty of 95 percent.

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2.13 Maintainability. Maintenance provisions shall be completed in accordance with paragraphs 3.14.1, 3.14.2, 3.14.3, and 4.5.7.1.1 of MIL-PRF-28800. The test set manufacturer shall provide all maintenance manuals, procedures, equipment, instructions, drawings, electrical schematics, etc. required for maintenance, troubleshooting, and repair of test set. The test set shall maximize use of automated self-test functions to identify failed components requiring replacement.

2.14 Automatic test mode. The test set shall provide an automated feature for conducting radio and beacon tests described in the commercial manuals and referenced technical orders (See 6.2.2). The test set shall provide the capability for full functional testing of each radio or beacon to verify proper operation. In the automatic mode, a test list shall be invoked upon operator selection of a particular radio or beacon. Upon initiation or start of test, the test set shall automatically cycle through the individual tests of the test list for the specified radio or beacon. The test set shall have a data display and capture capability for viewing of information, collection, storage, and printing of test results on an external printer. Information displayed and captured shall include identification of individual test, threshold limits, measured value, units, test conditions, and pass/fail result. In the event of a test failure, the test set shall provide options for failed item retest, failed item skip and continuation, debug, or test abortion. The automatic test function shall prompt the operator for inputs when required. The test set shall provide the operator the capability to set and change parameters (pass/fail criteria) for each test list without requiring software or hardware modifications.

2.15 Lists of test capabilities. The following lists of test capabilities (paragraphs 2.15.1 to 2.15.3) shall be included with the specified radio or beacon. Default test limits to be supplied with the test set and general test descriptions of the test parameters are included in 2.21 to 2.23. Specific test conditions and interfaces required to perform these tests shall be identified and provided by the manufacturer.

2.15.1 AN/PRC-112 radio test list. The following parameters shall be tested in the full functional test mode: Direct Current (DC) measurement, transmitter (TX) Radio Frequency (RF) power, TX frequency error, TX modulation, TX Beacon (BCN) frequency, TX BCN modulation, receiver (RX) audio level, RX sensitivity, RX distortion, RX speaker output, distance measuring equipment (DME) light emitting diode (LED) check, DME level check, DME uplink code check, and DME phase shift keying (PSK) delay check.

2.15.2 AN/PRC-90-2 radio test list. The following parameters shall be tested in the full functional test mode: DC direct current measurement, TX RF power, TX BCN HI power, TX frequency error, TX modulation, TX beacon frequency, TX beacon modulation, RX audio output, RX sensitivity, RX distortion, and RX speaker output.

2.15.3 AN/URT-33 beacon test list. The following parameters shall be tested in the full functional test mode: DC direct current measurement, TX RF power, TX beacon frequency, and

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TX beacon modulation. The test set shall provide the capability for loading and changing of upper and lower limit thresholds for each parameter without requiring software or hardware modifications.

2.16 Individual test mode. In the individual test mode, the operator shall have the capability to select and run single tests from a selected radio or beacon test list (see 2.15 above). As with the automatic mode, test information shall be displayed and captured for viewing and retrieval. In the event of a test failure, the test set shall provide options for failed item retest, failed item skip and continuation, debug, or test abortion.

2.17 Debug mode. In the event of an automatic or individual mode test failure, the operator shall be prompted with options for failed item retest, failed item skip and continuation, debug, or test abortion. The test set shall have provisions for debug operations and trouble shooting procedures for problem identification. The test set shall include an integral set of equipment functions to allow debug operations.

2.18 Test limits and test conditions. The test set shall be configured with factory default settings for upper and lower threshold limits and applicable test conditions for each parameter of the test list (see 2.21 to 2.23). The applicable test condition and interfaces required to perform these tests shall be identified and provided by the manufacturer. The test set shall provide the capability for changing default settings without requiring software or hardware changes. The test set shall have a feature (password, key, etc.) to control access to this capability only to authorized operators. The capability to use factory or user-defined test limits and test conditions shall be provided in automated, individual, or debug testing.

2.19 Test functions. The test set shall incorporate the functions of many different test instruments to provide the capability to troubleshoot and verify proper operation of communications and electronic equipment. The test set shall provide the capability for the operator to select among these functions without the use of additional test equipment. The required functions of the test set are listed below:

2.19.1 Receiver. The test set shall have the capability to function as a receiver for use in signal analysis. The capability to receive free space and direct coupled input signals from 500 kHz to 1 GHz shall be provided. Minimum input power requirements are 65 Watts and 200 Watts, respectively. The receiver shall be capable of demodulating AM, FM, SSB, and PM signals while monitoring radio frequency, radio frequency error, signal strength, RF power, modulation level, and AF frequency. The receiver shall be capable of detecting unknown carriers with an autoscan or user defined frequency scan feature.

2.19.2 Radio frequency (RF) signal generator. The test set shall have the capability to function as a signal generator producing signals in the 500 kHz to 1 GHz range in 100 Hz steps. Modulation capabilities shall include AM, FM, and PM. The capability for analog and digital

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modulation sources shall be provided. The RF output level shall be adjustable from –137 to 0 dBm with a 0.1 dB resolution and scaled in dBm or mV units.

2.19.3 Duplex mode. The test set shall have the capability to function in a duplex mode providing simultaneous receiver and generator control and display. Receiver and RF signal generator functions shall be available in the duplex mode.

2.19.4 Meter functions. The following meter functions shall be provided by the test set. The operator shall have the capability to select among the meter functions to perform functional tests of electronic and communication equipment. Meter functions, information display, controls, shall be inherent to the test set and shall not require use of additional test equipment.

2.19.4.1 Audio frequency (AF) measurement. The test set shall have the capability for making AF measurements. This function shall include operator selectable corner frequency adjustments for high pass and low pass signal filtering and a peak hold setting for the measurement of peak signal.

2.19.4.2 Frequency error measurement. The test set shall have the capability for making frequency error measurements. The frequency error shall be the difference between the received signal and the test set setting provided by the operator.

2.19.4.3 Power meter. The test set shall have the capability to conduct meter functions which include peak, average, and continuous wave power measurements of single side band, amplitude modulated, and frequency modulated signals. This function shall provide a peak hold capability and an auto-range or operator defined range capability for setting of the lower and upper power settings. The capability for inputting external gain adjustments (dB loss or gain) values shall also be provided by this function.

2.19.4.4 Deviation meter. The test set shall have the capability to measure signal deviations in peak and RMS values.

2.19.4.5 Modulation meter. The test set shall have the capability for making measurement of amplitude signal modulation. This function shall provide an auto-range capability and allow the operator to set upper and lower threshold limits. This function shall include an indicator which informs operator that modulation is outside threshold limits.

2.19.4.6 Distortion meter. The test set shall have the capability for measuring signal distortion in AM, FM, and PM signals. This function shall allow the operator to set upper and lower threshold limits for signal distortion. An indicator will inform the operator when measured distortion is outside threshold limits. Distortion range shall be from 0.1 percent to 20.0 percent with a 0.1 percent resolution. Signal input frequency and level shall be from 600 Hz to 1400 Hz at 0.1 VRMS to 30.0 VRMS.

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2.19.4.7 Bit error rate meter. The test set shall have the capability for measurement of bit error rates for data streams from 100 bits to 100 kBits at a rate from 75 BPS up to 16 kBPS.

2.19.4.8 Digital multimeter. The test set shall have the capability to function as a digital multimeter for use in measuring Voltage (AC/DC), Resistance, and Current (AC/DC). The test set shall have the capability to measure 0 to 1000 VDC and 0 to 500.0 VRMS with a frequency range of 50 Hz to 20 kHz. The test set shall have the capability to measure 0 to 20 Megohms and 0 to 2 amps (AC/DC).

2.19.5 Spectrum analyzer. The test set shall have the capability to function as a spectrum analyzer. Functions of the spectrum analyzer shall include the capability to store, recall, peak hold, average, track and compare trace modes. Dual markers for frequency and amplitude measurements shall also be pro-vided. The frequency range shall span 500 kHz to 1 GHz in 100 Hz steps with scan widths from 1 kHz to 100 MHz/division with an accuracy of +/- 5 percent of scan width. The vertical range and resolution shall be 10 dB/division and 2 dB/division.

2.19.6 Oscilloscope. The test set shall have the capability to function as a digitized oscilloscope. The oscilloscope shall have 1 MHz bandwidth and 200 Volts maximum input voltage. The amplitude range and sweep rates shall be 2 mV to 50 Volts/division and 2 microseconds to 100 microseconds per division, respectively. Markers for signal timing and amplitude shall also be provided.

2.19.7 Function generator. The test set shall have the capability to function as a signal generator to generate sine, square, triangle, ramp, and pulse waveforms. Programmable features for audio functions, tone sequence, and swept tone shall be included. The capability for digital data generation from 75 BPS to 16 kBPS shall be provided with random, fixed, or operator defined data patterns from 100 bits to 100 kbits. The frequency range shall be10 Hz to 40 kHz with levels ranging from 1.0 mVRMS to 2.5 VRMS.

2.20 Features.

2.20.1 AN/PRC-112 loader. The test set shall have the capability to load A and B frequencies and ID code for mission usage without requiring the load device.

2.20.2 Software expandability. The capability for software modifications and upgrades shall be provided. This shall include the incorporation of the requirements specified in 2.15 to 2.18 for additional radios and beacons. Software modifications or upgrades shall not require opening or disassembly of the test set.

2.20.3 Hardware expandability. The test set shall include provisions for hardware upgrades. Expansion slots, removable/replaceable components, expandable memory, or other means may be used.

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2.21 – 2.23 Parameter limits for testing. The parameter limits will be supplied by the manufacturer as factory default settings. In addition to parameter limits, applicable test conditions and interfaces to perform the tests will be identified and provided by the manufacturer. Following are the parameter limits for testing the AN/PRC-112 radio, AN/PRC-90-2 radio, and the AN/URT-33 beacon:

2.21. AN/PRC-112 RADIO TESTS LIMITS AND DESCRIPTIONS

2.21.1 DC CURRENT MEASUREMENT TEST LIMITS

CHANNEL	UPPER LIMIT (Amps)	LOWER LIMIT (Amps)
243.0 BCN	0.6	0.3
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CHA	N/A	N/A
CHB	N/A	N/A
243	0.125	0.09

TEST DEFINITION: TEST CHECKS POWER CONSUMPTION FOR OPERATION WITHIN SPECIFIED LIMITS

2.21.2. TRANSMITTER RF POWER TEST LIMITS

CHANNEL	UPPER LIMIT (mW)	LOWER LIMIT (mW)
243.0 BCN	2000	320
121.5 BCN	1000	65
121.5	510	32
282.8	800	125
CHA	800	125
CHB	800	125
243	800	125

TEST DEFINITION: TEST CHECKS THAT OUTPUT AMPLIFIER IS OPERATIONAL FOR ALL REQUIRED MODULATIONS. AVERAGE READINGS ARE TAKEN ON AM VOICE CHANNELS, AND PEAK READINGS ARE TAKEN FOR SWEPT TONE CHANNELS. SWITCH OPERATION ALSO VERIFIED ON 243 CHANNEL. ABOVE MEASUREMENTS ARE SPECIFIED TO ANTENNA ADAPTER.

2.21.3. TRANSMITTER FREQUENCY ERROR TEST LIMITS

CHANNEL	UPPER LIMIT (Hz)	LOWER LIMIT (Hz)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	500 (UHF)	-500 (UHF)
282.8	1200 (VHF)	1200 (VHF)
CH A	1200 (VHF)	1200 (VHF)
CH B	1200 (VHF)	1200 (VHF)
243	1200 (VHF)	1200 (VHF)

TEST DEFINITION: TEST CHECKS THAT TRANSMITTER CARRIER IS WITHIN SPECIFIED LIMITS ON ALL NON-BEACON CHANNELS.

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2.21. AN/PRC-112 RADIO TESTS LIMITS AND DESCRIPTIONS (continued)

2.21.4 TRANSMITTER MODULATION TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	100	70
282.8	100	70
CH A	100	70
CH B	100	70
243	100	70

TEST DEFINITION: TEST CHECKS THAT MODULATION IS WITHIN SPECIFIED LIMITS ON ALL NON-BEACON CHANNELS. A 1 kHz, 0.2 VRMS DRIVE SIGNAL IS APPLIED TO RADIO MICROPHONE TO GENERATE MEASURED SIGNAL.

2.21.5 TRANSMITTER BEACON FREQUENCY TEST LIMITS

CHANNEL	UPPER LIMIT (Hz)	LOWER LIMIT (Hz)
243.0 BCN	1600	300
121.5 BCN	1600	300
121.5	N/A	N/A
282.8	N/A	N/A
CH A	N/A	N/A
CH B	N/A	N/A
243	N/A	N/A

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR FREQUENCIES OCCURRING AT 300 TO 1600 Hz, 700 Hz SPAN MINIMUM

2.21.6. TRANSMITTER BEACON MODULATION TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	>80
121.5 BCN	N/A	>80
121.5	N/A	N/A
282.8	N/A	N/A
CH A	N/A	N/A
CH B	N/A	N/A
243	N/A	N/A

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR CORRECT MODULATION LEVEL;. SWEPT TONE DATA IS TAKEN TO OBTAIN 150 MODULATION MEASUREMENTS. MINIMUM MODULATION IS >80%.

2.21.7. RECEIVER AUDIO OUTPUT LEVEL TEST LIMITS

CHANNEL	UPPER LIMIT (mW)	LOWER LIMIT (mW)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	50	17
282.8	50	17
CH A	50	17
CH B	50	17
243	50	17

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2.21. AN/PRC-112 RADIO TEST LIMITS AND DESCRIPTIONS (continued)

TEST DEFINITION: TEST CHECKS FOR PROPER DRIVE SIGNAL FROM THE INTERNAL AUDIO AMP-LIFIER OF THE RADIO WITH VOLUME SET AT MAXIMUM. MEASURE-MENTS ARE TAKEN AT THE RADIO EARPHONE. APPLIED RF IS 30% MODULATED AT –33 dBm.

2.21.8 RECEIVER SENSITIVITY (1 kHz TONE) TEST LIMITS

CHANNEL	UPPER LIMIT	LOWER LIMIT (dB)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	>10
282.8	N/A	>10
CH A	N/A	>10
CH B	N/A	>10
243	N/A	>10

TEST DEFINITION: TEST CHECKS RADIO RECEPTION OF LOW LEVEL RF SIGNALS. APPLIED RF IS -87 dBm VHF AND -93 dBm UHF AT 30 % MODULATION WITH RADIO AUDIO FREQUENCY AT 0.5 VRMS. OUTPUT MEASUREMENT TAKEN AT RADIO EARPHONE.

2.21.9 RECEIVER DISTORTION (1 kHz TONE) TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	15	1
282.8	15	1
CHA	15	1
CH B	15	1
243	15	1

TEST DEFINITION: TEST CHECKS RADIO DISTORTION AT EARPHONE CONNECTOR; RADIO VOLUME AT 1.5 VRMS. APPLIED RF IS -33 dBm; -80 dBm at 30 % MODULATION.

2.21.10 RECEIVER SPEAKER OUTPUT

CHANNEL	UPPER LIMIT	LOWER LIMIT
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CHA	N/A	N/A
CH B	N/A	N/A
243	USER CHECK	USER CHECK

TEST DEFINITION: USER CHECKS FOR SPEAKER OPERATION USING A RF SIGNAL. A 1 kHz, 30 % MODULATION IS APPLIED TO A HIGH LEVEL RF SIGNAL AT -20 dBm ON A VOICE CHANNEL. USER LISTENS FOR CLEAR AND NOISE FREE SIGNAL.

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2.21. AN/PRC-112 RADIO TEST LIMITS AND DESCRIPTIONS (continued)

2.21.11 DME LED TEST LIMITS

CHANNEL	UPPER LIMIT	LOWER LIMIT
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CH A	USER CHECK	USER CHECK
CH B	N/A	N/A
243	N/A	N/A

TEST DEFINITION: THE RADIO IS LOADED WITH ID CODE AND DUPLEX FREQUENCY (251.9 MHz) IN CHANNEL A AND B. RADIO IS INTERROGATED AND USER CHECKS IF THE DME LED INDICATOR IS FLASHING.

2.21.12 DME MTL TEST LIMITS

CHANNEL	UPPER LIMIT	LOWER LIMIT
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CHA	3 CHECKS	3 CHECKS
CH B	N/A	N/A
243	N/A	N/A

TEST DEFINITION: THIS TEST CHECKS THE SENSITIVITY OF THE RADIO ON CHANNEL A FREQUENCIES: 227.55, 236, 243, 260.3, 270.3, AND 282.8 MHz. INTERROGATION OF RADIO IS PERFORMED USING A -86 dBm RF SIGNAL. INTERROGATION FOR EACH FREQUENCY IS PERFORMED 3 TIMES. SENSITIVITY IS ACCEPTABLE IF RADIO RESPONDS TO INTERROGATION.

2.21.13 DME UPLINK CODE CHECK TEST LIMITS

CHANNEL	UPPER LIMIT(%)	LOWER LIMIT (%)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CHA	DIGIT CHECKED	DIGIT CHECKED
CH B	N/A	N/A
243	N/A	N/A

TEST DEFINITION: THIS TEST CHECKS FOR TRANSMISSION OF PROPER UPLINK CODE FROM THE RADIO (THE LAST DIGIT ON THE RADIO ID CODE). TRANSMISSION OF RADIO SIGNAL IS RECEIVED AND DECODED; UPLINK CODE IS COMPARED TO STORED UPLINK CODE. CHECK IS PERFORMED 3 TIMES.

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2.21. AN/PRC-112 RADIO TEST LIMITS AND DESCRIPTIONS (CONTINUED)

2.21.14 DME PSK DELAY CHECK TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	N/A
121.5 BCN	N/A	N/A
121.5	N/A	N/A
282.8	N/A	N/A
CHA	4 STD DEV	0 STD DEV
CHB	N/A	N/A
243	N/A	N/A

TEST DEFINITION: A TIME MEASUREMENT IS PERFORMED OF ALL PSK; TRANSMIT DATA EDGES AS COMPARED TO THE PSK; RECEIVED DATA EDGES OF THE DME BURST. EDGE; DIFFERENCES ARE COLLECTED AND THE MEAN AND STANDARD DEVIATION ARE CALCULATED. CHECK IS PERFORMED A MINIMUM OF 25 TIMES AND ALL CHECKS MUST PASS. A SMALL STD DEV INDICATES CONSISTENT PHASE OFFSET BETWEEN TRANSMIT AND RECEIVED DME BURST.

2.22. AN/PRC-90-2 RADIO TEST LIMITS AND DESCRIPTIONS

2.22.1. DC CURRENT MEASUREMENT TEST LIMITS

CHANNEL	UPPER LIMIT (Amps)	LOWER LIMIT (Amps)
243.0 BCN	0.15	0.01
243	0.15	0.01
282.8	0.15	0.01

TEST DEFINITION: TEST CHECKS POWER CONSUMPTION FOR OPERATION WITHIN SPECIFIED LIMITS. TRANSMITTER MODULATION LEVEL SET AT 0.06 VRMS.

2.22.2. TRANSMITTER RF POWER TEST LIMITS

CHANNEL	UPPER LIMIT (mW)	LOWER LIMIT (mW)
243.0 BCN	1000	200
243	300	100
282.8	300	50

TEST DEFINITION: TEST CHECKS THAT OUTPUT AMPLIFIER POWER IS WITHIN SPECIFIED POWER RANGE. AVERAGE READINGS ARE TAKEN ON AM VOICE CHANNELS AND PEAK READINGS ARE TAKEN FOR SWEPT TONE CHANNELS.

2.22.3 TRANSMITTER BCN HI POWER TEST LIMITS

CHANNEL	UPPER LIMIT (mV\/)	LOWER LIMIT (mV\I)
243.0 BCN	1500	200
243	N/A	N/A
282.8	N/A	N/A

TEST DEFINITION: TEST CHECKS PEAK POWER OUTPUT WHEN ACTIVATING BCN SWITCH.

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2.22. AN/PRC-90-2 RADIO TEST LIMITS AND DESCRIPTIONS (continued)

2.22.4 TRANSMITTER FREQUENCY ERROR TEST LIMITS

CHANNEL	UPPER LIMIT (Hz)	LOWER LIMIT (Hz)
243.0 BCN	N/A	N/A
243	15000	-15000
282.8	15000	-15000

TEST DEFINITION: TEST CHECKS FREQUENCY ON VOICE CHANNELS.

2.22.5 TRANSMITTER MODULATION TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	N/A
243	100	70
282.8	100	70

TEST DEFINITION: TEST CHECKS THAT MODULATION IS WITHIN SPECIFIED LIMITS ON ALL NON-BEACON CHANNELS. DRIVE SIGNAL IS APPLIED TO RADIO MICROPHONE TO GENERATED MEASURED SIGNAL. TRANSMITTER MODULATION TONE AND LEVELS ARE 1 kHz AND 0.28 VRMS.

2.22.6 TRANSMITTER BEACON FREQUENCY TEST LIMITS

CHANNEL	UPPER LIMIT (Hz)	LOWER LIMIT (Hz)
243.0 BCN	1600	300
243	N/A	N/A
282.8	N/A	N/A

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR SWEEP OCCURRING BETWEEN 300 TO 1600 Hz, 700 Hz SPAN MINIMUM.

2.22.7. TRANSMITTER BEACON MODULATION TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	>80
243	N/A	N/A
282.8	N/A	N/A

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR CORRECT MODULATION LEVEL. SWEPT TONE DATA IS TAKEN TO OBTAIN 150 MODULATION MEASUREMENTS. MIN MODULATION IS >80%.

2.22.8 RECEIVER AUDIO OUTPUT TEST LIMITS

CHANNEL	UPPER LIMIT (mW)	LOWER LIMIT (mW)
243.0 BCN	N/A	N/A
243	30	15
282.8	30	15

TEST DEFINITION: TEST CHECKS OUTPUT LEVEL AT RADIO EARPHONE. APPLIED RF IS -75 dBm WITH 30 % MODULATION.

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2.22. AN/PRC-90-2 RADIO TESTS LIMITS AND DESCRIPTIONS (continued)

2.22.9 RECEIVER SENSITIVITY TEST LIMITS

CHANNEL	UPPER LIMIT (dB)	LOWER LIMIT (dB)
243.0 BCN	N/A	N/A
243	40	12
282.8	40	12

TEST DEFINITION: TEST CHECKS RADIO RECEPTION OF LOW LEVEL RF SIGNALS. APPLIED RF IS -81 dBm WITH 30 % MODULATION. MEASUREMENT TAKEN AT RADIO EARPHONE. RADIO AUDIO LEVEL SET AT 0.5 VRMS.

2.22.10 RECEIVER DISTORTION TEST LIMITS

CHANNEL	UPPER LIMIT	LOWER LIMIT
243.0 BCN	N/A	N/A
243	10	1
282.8	10	1

TEST DEFINITION: TEST CHECKS RADIO DISTORTION AT EARPHONE CONNECTOR APPLIED RF IS -73 dBm AT 1 kHz, 30% MODULATION. RADIO AUDIO LEVEL SET AT 0.5 VRMS.

2.22.11 RECEIVER SPEAKER OUTPUT TEST LIMITS

CHANNEL	UPPER LIMIT	LOWER LIMIT
243.0 BCN	N/A	N/A
243	N/A	N/A
282.8	USER CHECK	USER CHECK

TEST DEFINITION: USER CHECKS FOR SPEAKER OPERATION USING AN RF SIGNAL. APPLIED RF IS -73 dBm WITH 1 kHz, 30 % MODULATED TONE. USER LISTENS FOR CLEAR AND NOISE FREE SIGNAL.

2.23. AN/URT-33 BEACON TESTS LIMITS AND DESCRIPTIONS

2.23.1 DC CURRENT MEASUREMENT TEST LIMITS

CHANNEL	UPPER LIMIT (Amps)	LOWER LIMIT (Amps)
243.0 BCN	0.25	0.05

TEST DEFINITION: TEST CHECKS POWER CONSUMPTION FOR OPERATION WITHIN LIMITS.

2.23.2 TRANSMITTER RF POWER TEST LIMITS

CHANNEL	UPPER LIMIT (mW)	LOWER LIMIT (mW)
243.0 BCN	777	389

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2.23. AN/URT-33 BEACON TESTS LIMITS AND DESCRIPTIONS (continued)

TEST DEFINITION: TEST CHECKS THAT OUTPUT AMPLIFIER POWER IS WITHIN SPECIFIED POWER RANGE. PEAK READINGS ARE TAKEN FOR SWEPT TONE MODULATIONS AT 243 MHz.

2.23.3 TRANSMITTER BEACON FREQUENCY TEST LIMITS

CHANNEL	UPPER LIMIT (Hz)	LOWER LIMIT (Hz)
243.0 BCN	1600	300

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR SWEEP OCCURRING BETWEEN 300 TO 1600 Hz, 700 Hz SPAN MINIMUM.

2.23.4 TRANSMITTER BEACON MODULATION TEST LIMITS

CHANNEL	UPPER LIMIT (%)	LOWER LIMIT (%)
243.0 BCN	N/A	>80

TEST DEFINITION: TEST CHECKS BEACON CHANNEL DEMODULATED SWEPT TONE FOR CORRECT MODULATION LEVEL. SWEPT TONE DATA IS TAKEN TO OBTAIN 150 MODULATION MEASUREMENTS. MINIMUM MODULATION IS >80%.

3. REGULATORY PROVISIONS.

3.1 Recoverable materials. In accordance with Section 23.403 of the Federal Acquisition Regulation (FAR), the Federal Government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The contractor shall certify and maintain substantiating evidence that the test set offered meets the salient characteristics of this CID, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial market. The government requires proof of such conformance prior to first delivery and thereafter, as may be otherwise provided under the provisions of the purchase order or contract. Information substantiating product compliance with the requirements of this CID shall be submitted concurrently with the bid sample unless otherwise specified in the purchase order or contract.

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4.2 Bid sample(s). The offeror shall submit a bid sample for government test and evaluation in accordance with the purchase order or contract. Government evaluation will consist of inspections, demonstrations, and tests to assess compliance with the requirements herein and to determine whether the offered test set provides accurate and repeatable measurements, and has acceptable form, fit, function and suitability for use.

5. PACKAGING.

5.1 Preservation, packing, and packaging requirements for this test set shall be as specified in the contract or purchase order.

6. NOTES.

6.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables contained in the latest version of FED-STD-376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2 Source of documents.

6.2.1 Document ordering. Copies of federal and Department of Defense specifications, standards, and handbooks are available from the Defense Automated Printing Service, (DAPS), 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Phone 215/697-2197.

6.2.2 Technical Manuals. Copies of technical and commercial manuals may be obtained from: 311 HSW/YACL, 8107 13th Street, Brooks AFB, TX 78235.

6.2.2.1 Army Technical Manual TM 11-5820-1037-13&P
6.2.2.2 Navy Manual NAVAIR 16-35PRC112-1
6.2.2.3 Air Force Technical Manual TO 31R2-2PRC-112-1
6.2.2.4 Air Force Technical Manual TO 31R4-2URT33-11
6.2.2.5 Air Force Technical Order TO 31R2-2PRC90-1
6.2.2.6 Air Force Technical Order TO 31R2-2PRC90-2.

6.3 Ordering data. The following information should be provided at the time of order:

a. Title, CID document number and date.
b. Product samples required. (see 4.2).
c. Packaging requirements (see 5.1).
d. Test lists and parameter limits (see 2.21 through 2.23.)

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6.4 Possible acceptable products. The following commercial products may meet the salient characteristics of this CID:

TABLE I. Possible acceptable products.

Part Number	Manufacturer	CAGE Code	National Stock Number
RCTS-03	IFR Systems, Inc.	51190	6625-01-457-1265

MILITARY INTEREST:

Custodian:
Air Force - 99

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Air Force - 82

Agent:
Air Force - 99

Project 6625-F077

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A--59407	**2. DOCUMENT DATE (YYMMDD)** February 26, 1999
3. DOCUMENT TITLE TEST SET GROUP, RADIO		
4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*		
5. REASON FOR RECOMMENDATION		
6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**
8. PREPARING ACTIVITY		
a.	**b. TELEPHONE** *(Include Area Code)* (1) Commercial: (2) DSN	
c.	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340 DSN 289-2340**	

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